Digital Egypt Pioneers Initiative (DEPI)
Logistics Performance Analysis

Capstone Project — CAI4_DAT2_S1 [R4]
Track: Data Analytics
Major: Microsoft Power BI Specialist

Instructor: Mahmoud Seraj

Project Overview & Objectives

Project Objective

This project analyzes two full years of delivery operations data spanning January 2023 through December 2024 for a logistics network operating across six distribution hubs. The dataset covers 27,979 individual delivery orders, involving 55 drivers, 45 vehicles, and six hub locations.

Using Microsoft Power BI as the primary analytics platform, the project transforms raw operational data into a suite of interactive dashboards designed to give decision-makers at all levels a clear, real-time view of network performance and improvement opportunities.

Central Challenges

- Delay rate of 21.12% across all delivered orders
- Aging vehicle fleet contributing to frequent breakdowns
- Uneven workload distribution across hubs
- Absence of centralized system for tracking driver-level performance

27,979

Total Orders

Across all six hubs, January 2023 – December 2024

21.12%

Delay Rate

5,908 orders delivered late across the two-year period

84.4%

CSAT Score

Percentage of orders receiving a score of 4 or 5

73.33%

Active Vehicles

33 out of 45 vehicles available for dispatch

Project Methodology

01

Data Preparation

Four source CSV files imported and processed using Power Query. Standardized date formats, renamed columns for consistency, handled null values, and created calculated columns: Vehicle Age, Driver Tenure, and Experience Segment.

02

Data Modeling

Constructed star schema with Orders table at center as fact table. Drivers, Hubs, and Vehicles tables connected as dimension tables via matching keys. Ensures filters propagate correctly across all visuals.

03

Exploratory Data Analysis

Studied relationships between driver experience and delivery performance, analyzed vehicle age versus breakdown frequency, examined delivery time effects on customer satisfaction, and identified peak order months.

04

Dashboard Development

Built nine interactive report pages in Power BI, each targeting specific audience and analytical questions. All pages include slicers for Year, Month, Hub, Vehicle Type, and Driver Name with MoM comparisons.

Stakeholder Analysis

Understanding who uses the dashboards and what metrics matter most to their decision-making:

Stakeholder	Role	Key Metrics Needed
Logistics Operations Manager	Strategic decision-maker	Network-wide KPIs, hub performance comparisons, monthly trends across full two-year period
Hub Supervisors	Operational management	Hub processing time and correlation with CSAT, ranking, workload rebalancing
Driver Manager	Driver performance oversight	Driver performance ratings, top delayed orders by driver name, individual driver drill-down profiles
Fleet Manager	Vehicle and maintenance oversight	Breakdown rates by vehicle model, current maintenance rate, active versus unavailabl

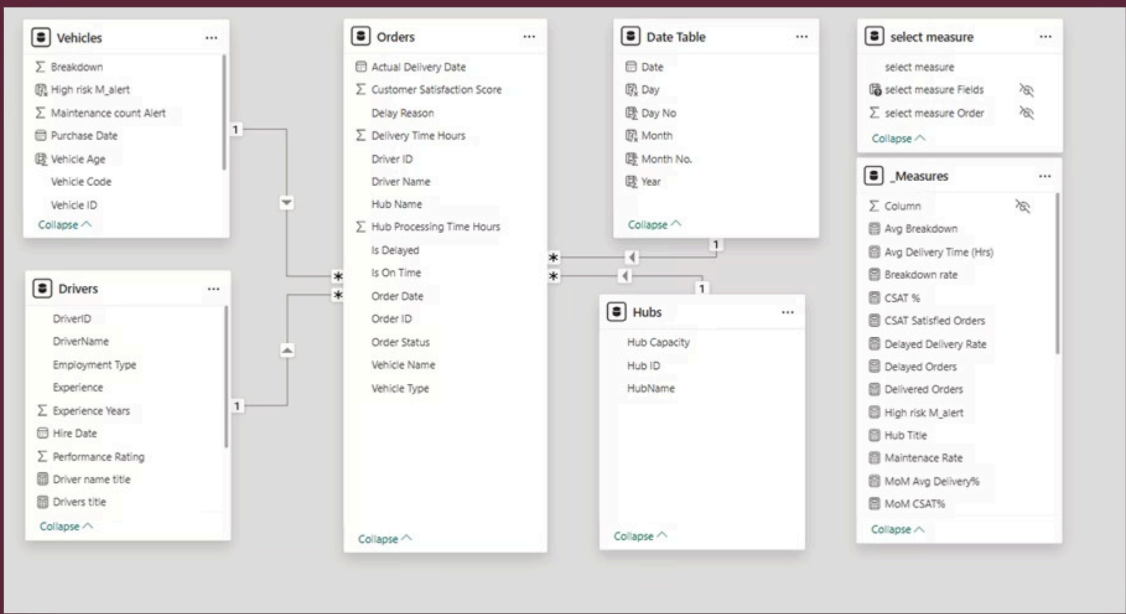
Dataset Overview & Structure

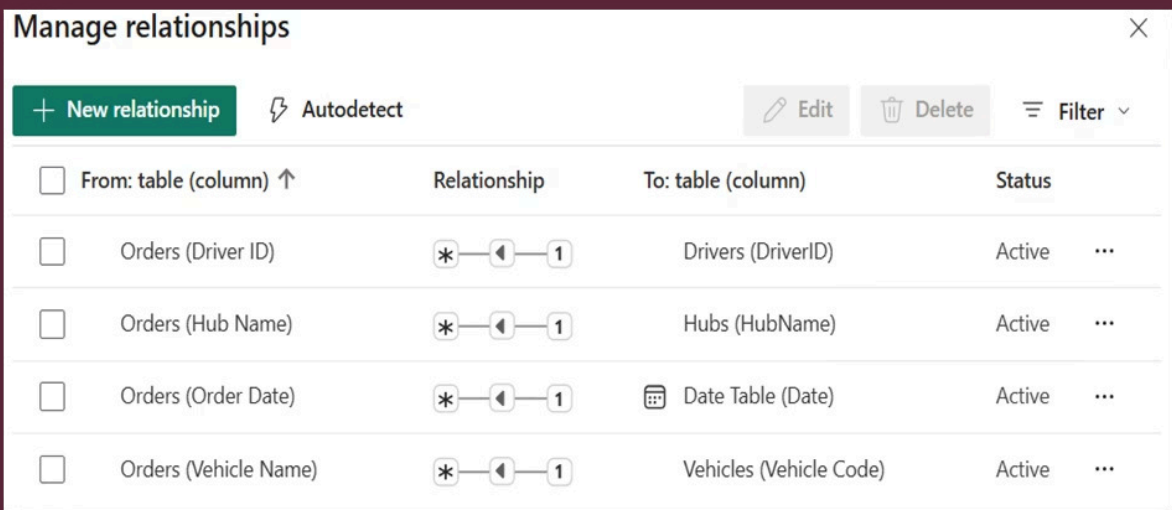
All data provided as CSV files covering January 2023 to December 2024. Dataset structured across four tables describing every aspect of logistics operations:

Table	Records	Description
Orders	27,979	Central fact table — one row per delivery order
Drivers	55	One row per driver with employment and performance data
Vehicles	45	One row per vehicle with status and breakdown history
Hubs	6	One row per hub with location and capacity

Data Modelling and Relationships:

The four tables are connected using a star schema. The Orders table serves as the central fact table, while Drivers, Vehicles, and Hubs act as dimension tables connected to it via their respective keys. This architecture ensures that any filter applied on the report — such as selecting a specific hub or a date range — automatically propagates to all visuals across the page.




Manage relationships

New relationship | Autodetect | Edit | Delete | Filter

From: table (column)	Relationship	To: table (column)	Status
Orders (Driver ID)	* — 1	Drivers (DriverID)	Active
Orders (Hub Name)	* — 1	Hubs (HubName)	Active
Orders (Order Date)	* — 1	Date Table (Date)	Active
Orders (Vehicle Name)	* — 1	Vehicles (Vehicle Code)	Active

Network-Wide Performance Metrics

On-Time Delivery Rate	Average Delivery Time	Customer Satisfaction
78.88% network-wide OTD Approximately 1 in 5 orders delayed	**35.78 hours** from order to delivery On-time: 35.78 hrs | Delayed: 42.48 hrs	**4.17/5** average score 84.4% of orders rated 4 or 5

Top Delay Causes

- **Road Construction:** 623 cases (external factor)
- **Vehicle Breakdown:** 623 cases (controllable through maintenance)
- Weather Conditions: 587 cases
- Hub Processing Delays: 542 cases

Vehicle Fleet Status

- **Total Vehicles:** 45 across 9 models
- **Active:** 33 (73.33%)
- **Under Maintenance:** 12 (26.67%)
- **Average Age:** 5.2 years

Overview Dashboard

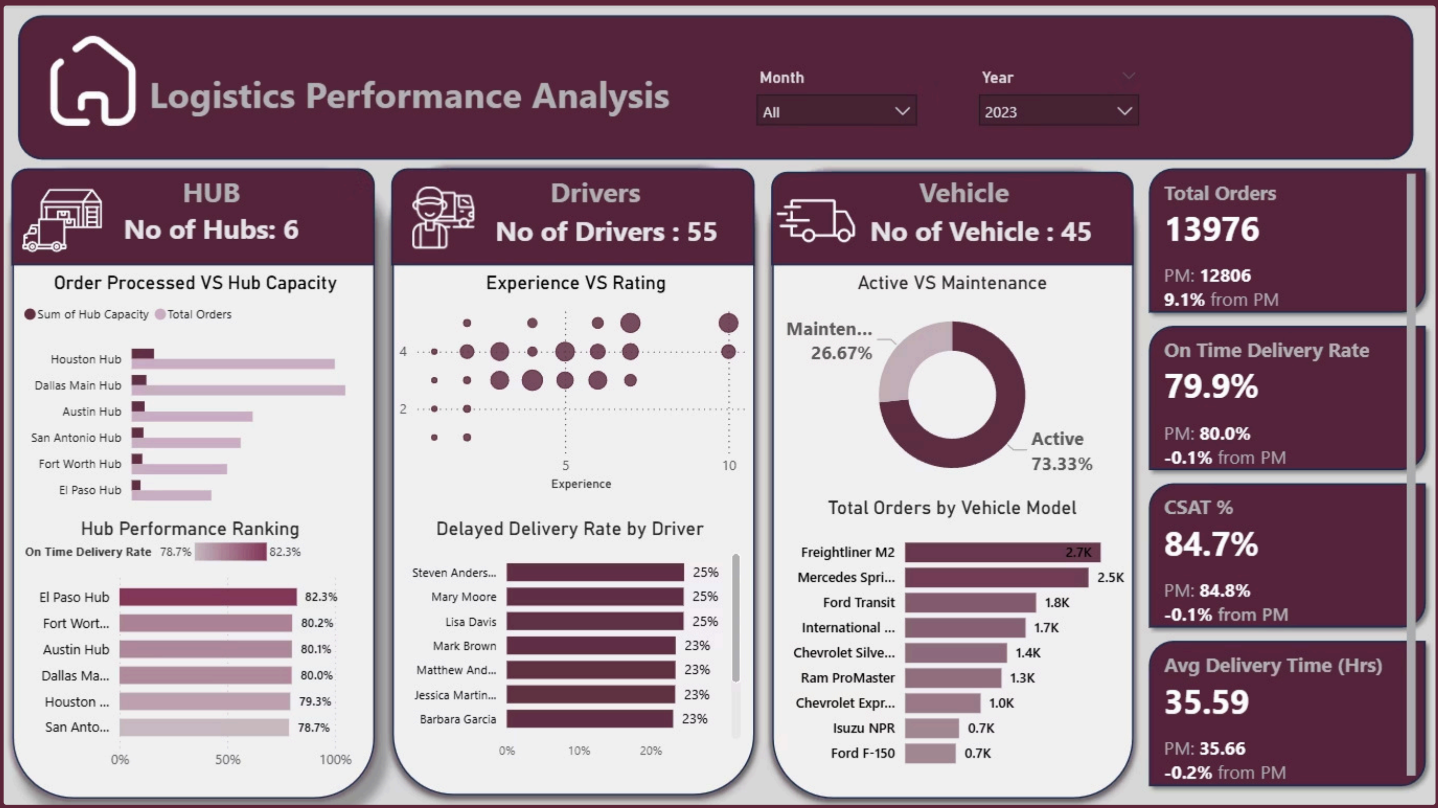
Hub utilization imbalance: Gap between hub capacity and actual orders, especially in Dallas and Houston, indicating inefficient resource use.

Best vs worst hub: El Paso has the highest on-time delivery rate (~81%), while Austin is the lowest (~78.7%), pointing to operational issues.

Driver performance variation: A few drivers (e.g., Mark Brown, Robert) have high delay rates (~24–25%), hurting overall efficiency.

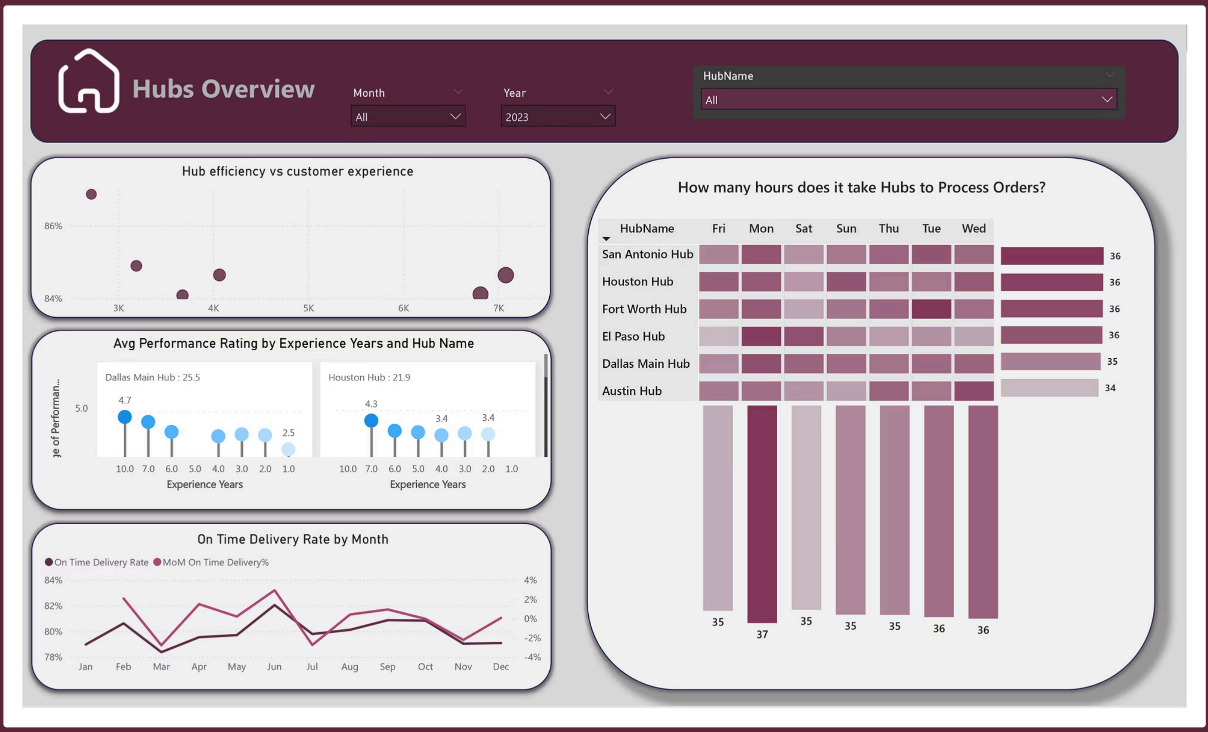
Around 73% of vehicles are active, while 26% are under maintenance.

This is a relatively high maintenance ratio and may affect delivery capacity and speed.



Hub Performance Analysis

How to read it: Start with the bar chart to see which hub has the highest absolute delay count. Move to the scatter to understand whether that hub's delays correlate with longer processing times. Use the matrix to identify which day of the week creates the worst bottleneck. The line chart at the bottom shows whether hub performance is improving or declining over time.



Why it matters: Hub-level delay data often reveals that individual driver delay counts are partially determined by the hub they are assigned to. Before attributing high delay counts to a driver, check whether other drivers at the same hub show similar patterns.

Key Finding: Hub processing time matrix reveals Wednesdays show consistently longer processing times at Fort Worth Hub (38 hours) compared to network's typical 31-37 hours, signaling mid-week capacity constraint.

Drivers Overview

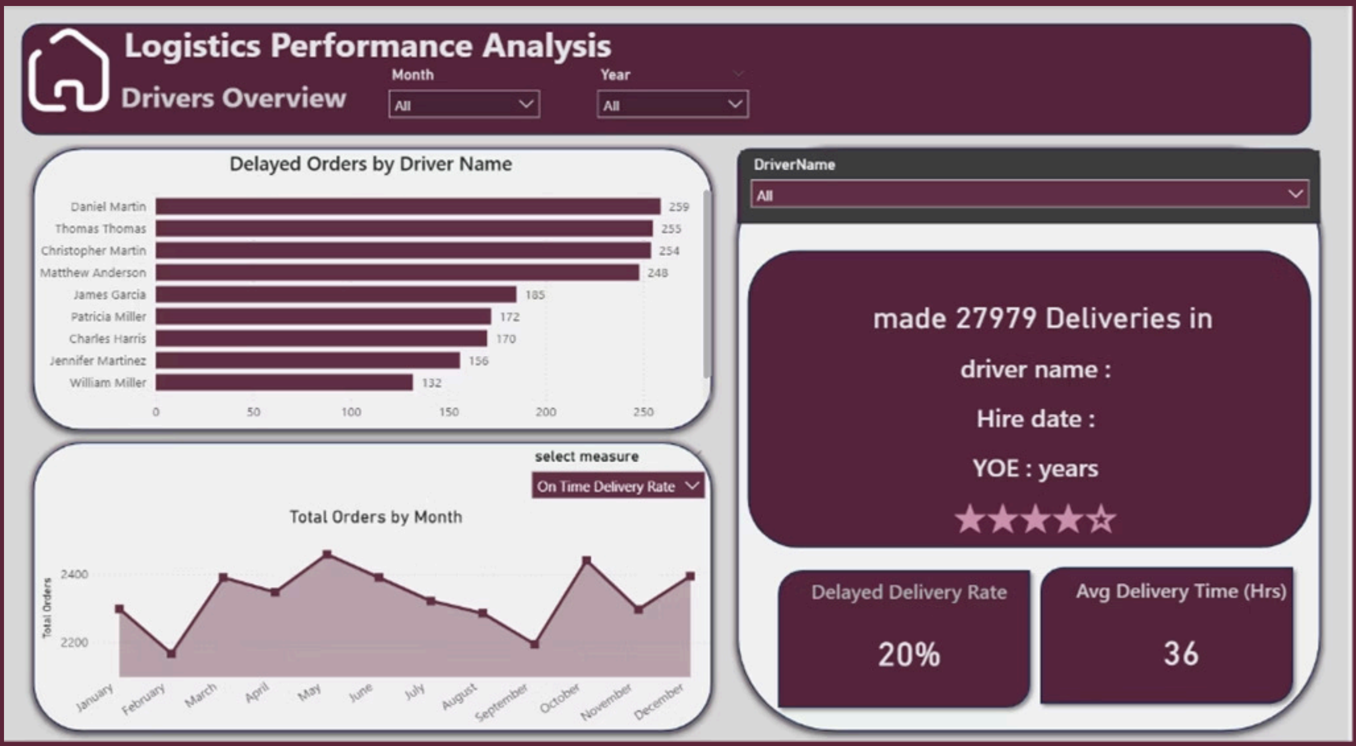
This section highlights individual driver profiles and performance metrics. It includes driver name, hire date, and years of experience (YOE), which help assess each driver's capability. The average delivery time is 36 hours, and the delayed delivery rate is 20%, indicating variation in efficiency and reliability across drivers.

Some drivers have higher numbers of delayed orders, with Daniel Martin recording the highest delays.

> ⓘ The order volume chart tracks total monthly orders to identify seasonal trends. It highlights peak activity periods, such as the surge in October, compared with lower-demand months like February and September. Understanding these fluctuations is essential for adjusting fleet capacity and ensuring delivery deadlines are met during high-demand periods.

NO FILTER APPLIED VS

YEAR 2023 FILTER APPLIED



Vehicles Overview

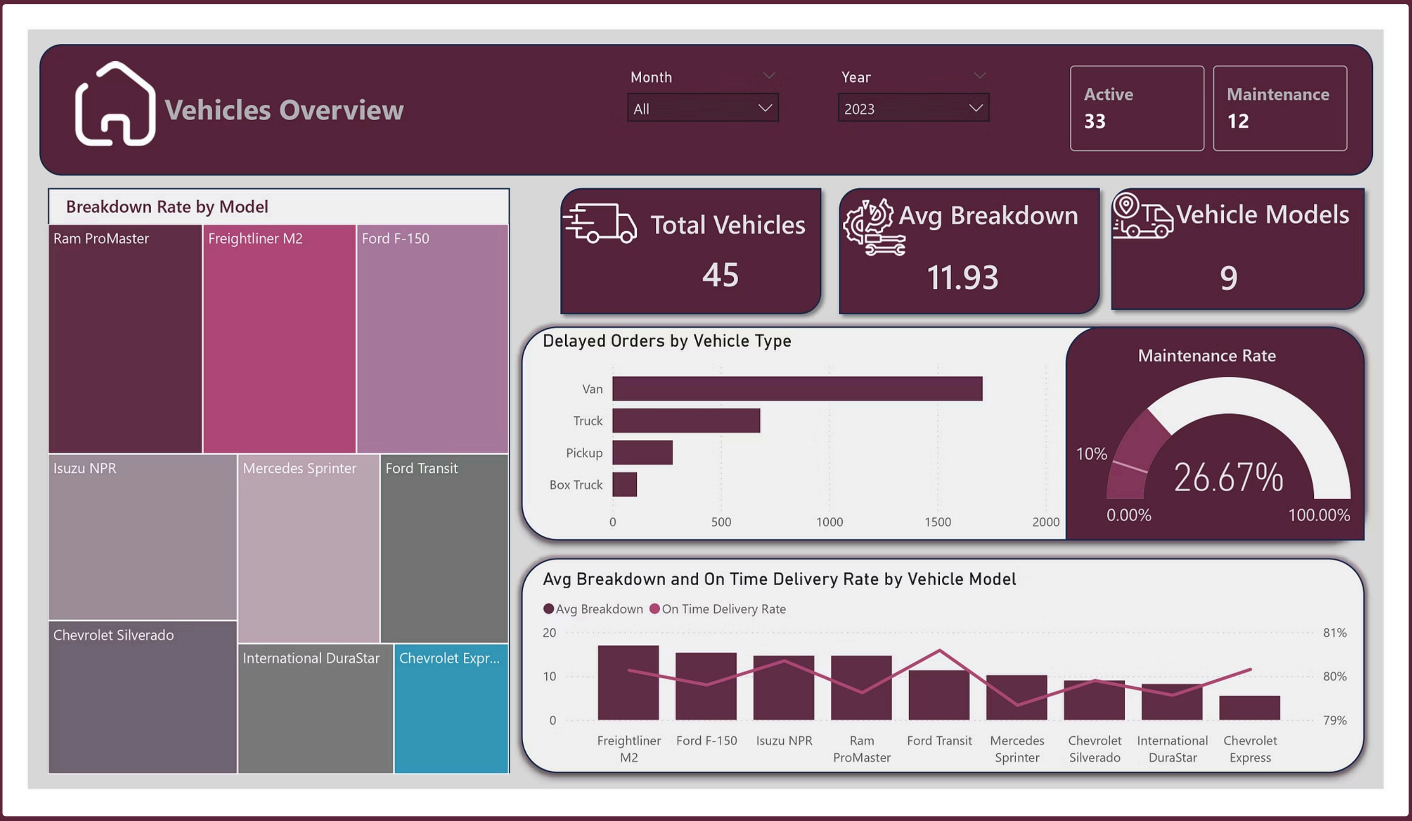
The company operates a fleet of 45 vehicles across 9 different models, providing flexibility in delivery operations.

About 26.67% of vehicles are under maintenance, which could reduce operational efficiency and increase delivery pressure on active vehicles.

Vans have the highest delayed order rate, indicating that this vehicle type may face heavier workload or operational challenges.

Breakdown rates differ between vehicle models, with some models showing higher maintenance issues than others.

Vehicle reliability has a direct impact on performance, as models with fewer breakdowns tend to achieve better on-time delivery rates.



Model	Count	Avg Breakdowns
Freightliner M2	9	17.0
Ford F-150	3	15.3
Isuzu NPR	3	14.7
Ram ProMaster	3	14.7
Mercedes Sprinter	9	10.2
Chevrolet Express	4	5.5

DAX Metrics: Order Volume Calculations

These measures track order volume across the network and support month-over-month analysis.

Total Orders

```
Total Orders = COUNT(Orders[Order ID])
```

Business Meaning: Counts all orders in the dataset, including delivered, delayed, and cancelled orders.

PM Orders (Previous Month Orders)

```
PM Orders = CALCULATE([Total Orders], DATEADD('Date Table'[Date], -1, MONTH))
```

Business Meaning: Returns the total order count for the previous calendar month. It serves as the comparison base for MoM calculations.

MoM Orders %

```
MoM Orders % = DIVIDE([Total Orders] - [PM Orders], [PM Orders])
```

Business Meaning: Shows the month-over-month change in order volume as a percentage. Positive values indicate growth; negative values indicate decline.

Delivered Orders

```
Delivered Orders = CALCULATE([Total Orders], Orders[Order Status] = "Delivered")
```

Business Meaning: Counts only orders marked as delivered. Use this as the base for delivery performance metrics.

Date Table Enhancements

Two calculated columns enable day-of-week analysis in hub processing time matrices.

Day (Calculated Column)

```
Day = FORMAT('Date Table'[Date], "ddd")
```

Business Meaning: Formats each date as a three-letter day abbreviation (Mon, Tue, Wed, etc.) for matrix column headers.

Day No (Calculated Column)

```
Day No = WEEKDAY('Date Table'[Date], 2)
```

Business Meaning: Assigns numeric values (1 = Monday through 7 = Sunday) to enable correct weekday sorting instead of alphabetical order.

Delivery Performance Metrics

These measures track delivery timeliness and month-over-month performance.

On Time Delivered Orders

```
On Time Delivered Orders = CALCULATE([Total Orders], Orders[Delivery Status] = "On Time")
```

Business Meaning: Counts orders delivered on time.

On Time Delivery Rate (OTD%)

```
On Time Delivery Rate (OTD%) = DIVIDE([On Time Delivered Orders], [Delivered Orders])
```

Business Meaning: Measures the share of delivered orders that arrive on time.

PM On Time Delivery Rate

```
PM On Time Delivery Rate = CALCULATE([On Time Delivery Rate (OTD%)], DATEADD('Date Table'[Date],
-1, MONTH))
```

Business Meaning: Returns the prior month's on-time delivery rate for comparison.

MoM On Time Delivery %

```
MoM On Time Delivery % = DIVIDE([On Time Delivery Rate (OTD%)] - [PM On Time Delivery Rate], [PM
On Time Delivery Rate])
```

Business Meaning: Shows the month-over-month change in on-time delivery rate.

CSAT Metrics

These measures track customer satisfaction and month-over-month performance.

CSAT Satisfied Orders

```
CSAT Satisfied Orders = CALCULATE([Total Orders], Orders[CSAT Status] = "Satisfied")
```

Business Meaning: Counts orders where customers reported being satisfied.

CSAT %

```
CSAT % = DIVIDE([CSAT Satisfied Orders], [Total Orders])
```

Business Meaning: Measures the share of total orders that received a satisfied CSAT response.

PM CSAT

```
PM CSAT = CALCULATE([CSAT %], DATEADD('Date Table'[Date], -1, MONTH))
```

Business Meaning: Returns the prior month's CSAT percentage for comparison.

MoM CSAT %

```
MoM CSAT % = DIVIDE([CSAT %] - [PM CSAT], [PM CSAT])
```

Business Meaning: Shows the month-over-month change in CSAT percentage.

Time Metrics

These measures track delivery duration and how it changes over time.

Avg Delivery Time (Hrs)

```
Avg Delivery Time (Hrs) = AVERAGE(Orders[Delivery Time Hours])
```

Business Meaning: Calculates the average hours from order placement to delivery. Lower values indicate faster fulfillment.

PM Avg Delivery Time

```
PM Avg Delivery Time = CALCULATE([Avg Delivery Time (Hrs)], DATEADD('Date Table'[Date], -1, MONTH))
```

Business Meaning: Returns the prior month's average delivery time for comparison.

MoM Avg Delivery %

```
MoM Avg Delivery % = DIVIDE([Avg Delivery Time (Hrs)] - [PM Avg Delivery Time], [PM Avg Delivery Time])
```

Business Meaning: Shows the percentage change in delivery time versus prior month. Positive values indicate slower delivery (unfavorable).

Year Month Title

```
Year Month Title = SELECTEDVALUE('Date Table'[Month]) & " " & SELECTEDVALUE('Date Table'[Year])
```

Business Meaning: Generates a dynamic text title showing the selected month and year (e.g., 'June 2

Vehicle Metrics

These measures track fleet health and identify vehicles requiring maintenance attention.

Total Vehicles

```
Total Vehicles = COUNT(Vehicles[Vehicle ID])
```

Business Meaning: Counts the total number of vehicles in the fleet (currently 45 across nine models).

Avg Breakdown

```
Avg Breakdown = AVERAGE(Vehicles[Breakdown])
```

Business Meaning: Calculates the average breakdowns per vehicle. Current fleet average is 11.93 breakdowns per vehicle.

Vehicle Age (Calculated Column)

```
Vehicle Age = YEAR(TODAY()) - YEAR(Vehicles[Purchase Date]) & " years"
```

Business Meaning: Computes each vehicle's current age in years. Shows 0.655 correlation with breakdown frequency.

Maintenance Rate

```
Maintenance Rate = DIVIDE(CALCULATE(COUNT(Vehicles[Vehicle ID]), Vehicles[Vehicle Status] = "Maintenance"),
COUNT(Vehicles[Vehicle ID]))
```

Business Meaning: The percentage of vehicles currently in maintenance status (currently 26.67%).

Breakdown Rate

```
Breakdown Rate = DIVIDE(SUM(Vehicles[Breakdown]), SUM(Vehicles[Vehicle Age]), 0)
```

Business Meaning: Measures the frequency of vehicle failures over time. Higher values indicate fleet reliability issues.

High Risk Alert

```
High Risk Alert = IF([Breakdown Rate] > 4, "High Risk", "Normal")
```

Business Meaning: Flags vehicles or fleet segments as "High Risk" when breakdown rate exceeds 4 incidents per year.

Dynamic Labels & Titles

These measures are designed to create dynamic text labels and titles for various visuals and page headers, enhancing the interactivity and readability of reports.

5.7 Hub Metrics

These measures generate dynamic text labels for hub-related visuals and page headers.

No of Hubs

```
No of Hubs = "No of Hubs: " & COUNT(Hubs[Hub ID])
```

Business Meaning: Creates a dynamic text string displaying the hub count (e.g., 'No of Hubs: 6') for summary cards.

Hub Title

```
Hub Title = "How many hours does it take Hubs to Process Orders?"
```

Business Meaning: Static text measure used as the dynamic title for the hub processing time matrix.

5.8 Driver Metrics

These measures generate dynamic text labels for driver-related visuals and page headers.

Star Rating

```
Star Rating = REPT(UNICHAR(9733), AVERAGE(Drivers[Performance Rating])) & REPT(UNICHAR(9734), 5 - AVERAGE(Drivers[Performance Rating]))
```

Business Meaning: Displays average driver performance rating as filled and empty stars (e.g., ★★★★☆).

No of Drivers

```
No of Drivers = "No of Drivers: " & COUNT(Drivers[DriverID])
```

Business Meaning: Creates a dynamic text string displaying the driver count (e.g., 'No of Drivers: 55') for summary cards.

Driver Title

```
Driver Title = SELECTEDVALUE(Drivers[DriverName]) & " made " & [Total Orders] & " Deliveries in " & SELECTEDVALUE('Date Table'[Month]) & " " & SELECTEDVALUE('Date Table'[Year])
```

Business Meaning: Dynamic title for driver overview showing selected driver's name, order count, and month/year.

YOE (Years of Experience)

```
YOE = "YOE: " & SELECTEDVALUE(Drivers[Experience Years]) & " years"
```

Business Meaning: Dynamic text label showing selected driver's years of experience in readable format.

Hire Date Title

```
Hire Date Title = "Hire date: " & SELECTEDVALUE(Drivers[Hire Date])
```

Business Meaning: Dynamic text label displaying the selected driver's hire date in user-friendly format.

Driver Name Title

```
Driver Name Title = "Driver name: " & SELECTEDVALUE(Drivers[DriverName])
```

Business Meaning: Dynamic text label displaying the selected driver's name in user-fri

Summary of Key Findings

1

One in Five Orders is Delayed

The network delay rate is 21.12%, meaning 5,908 out of 27,979 orders were delivered late during the two-year period. The two most frequent delay causes — Road Construction and Vehicle Breakdown — are tied at 623 cases each. However, Road Construction is an external factor while Vehicle Breakdown is entirely within the company's control through preventive maintenance and fleet renewal.

2

Customer Satisfaction is Strong but Sensitive to Delivery Speed

The network averages 4.17 out of 5 in customer satisfaction (85.3% CSAT rate). However, delayed orders take 42.48 hours to complete versus 35.78 hours for on-time orders — a 6.7-hour gap that is directly reflected in lower satisfaction scores. The correlation between delivery time and CSAT is −0.287, confirming that faster delivery consistently produces better customer experiences.

3

Hub Workload is Significantly Uneven

Dallas Main Hub processes 26.2% of all network orders — the highest volume of any hub — despite not holding the largest capacity in the network. El Paso Hub handles the fewest orders (2,718) and achieves the best OTD rate (80.6%), reinforcing the hypothesis that order pressure affects delivery performance. Rebalancing workload from high-pressure hubs to under-utilized ones could improve network-wide OTD rates.

4

Driver Experience Does Not Predict Performance

The correlation between experience years and delivery time or CSAT score is 0.006 — effectively zero. New drivers and senior drivers perform almost identically across all measured metrics. This means that current hiring or evaluation frameworks that weight experience heavily are not aligned with actual performance data.

5

Fleet Health is a Critical Operational Risk

26.67% of the fleet — 12 out of 45 vehicles — is unavailable for dispatch at any given time. The average breakdown count across the fleet is 11.93 per vehicle, and vehicle age is a meaningful predictor of breakdown frequency (correlation: 0.655). Freightliner M2 vehicles average 17.0 breakdowns each, while the oldest vehicles in the fleet are 8 years old. This level of fleet unreliability directly contributes to Vehicle Breakdown being one of the top two delay causes.

6

Contract Drivers Outperform All Other Employment Types

Despite representing only two drivers and 1,105 orders, contract drivers achieve an OTD rate of 80.6% and an average delivery time of 34.83 hours — the best figures of any employment category. Part-time drivers, by contrast, show the lowest OTD rate (77.9%) and the longest average delivery time (36.72 hours). These differences warrant further investigation into scheduling practices and route assignment.

Strategic Recommendations

Plan Ahead for Peak Months

May and October are highest-volume months every year. Hire additional contract drivers in advance and clear maintenance backlogs before volume surges to protect OTD rates and CSAT scores.

Shift Driver Evaluation to Outcome-Based Metrics

Experience years do not predict performance (correlation: 0.006). Evaluate drivers on actual OTD rate, delay count, and CSAT score over rolling six-month period instead of experience.

Implement Vehicle Retirement Policy

Place vehicles 7+ years old with 20+ breakdowns on immediate retirement schedule. Set preventive maintenance milestones at breakdown count thresholds rather than calendar intervals.

Rebalance Hub Workload

Route portion of Dallas Main Hub's volume to underutilized hubs — particularly Fort Worth and El Paso. El Paso's superior OTD rate at lower volumes suggests room to absorb additional orders.

Coach Using Cross-Referenced Evidence

Before assigning coaching to top 10 highest-delay drivers, cross-reference delay records against vehicle breakdown data. If delays correlate with vehicle history, problem may be fleet-related rather than driver-related.

Project Team:

Team Member	Contribution
Jana El Naggar	Data Cleaning, Vehicles Analysis Dashboard, Insights and Recommendations & Document.
Shimaa Samy	Overview Dashboard (Home Page) & Presentation.
Mariam Hesham	Hubs Analysis Dashboard & Document.
Mariam Mowafy	Drivers Overview Dashboard.
Fatma Ez-zahraa	Documentation.

Project Github Link: **Depi2026/Logistics-Project**